

Software Requirements Specification

Introduction

In competitive gaming, post-match analysis is often limited to video replays (VODs), which lack objective spatial data. Professional teams and analysts need a way to visualize positioning, rotations, and map control through coordinate-based data to identify patterns that are invisible to the naked eye.

Vantage Point will ingest game coordinate data, process spatial relationships, and provide a dashboard for analysing player positioning and team density over the course of a match.

User Stories & Characteristics

User Characteristics

The system has been set up to allow for three different kinds of users:

1. **Players:** A general user who will use the system to get feedback on League of Legends game-play for themselves and their team mates
2. **Admin:** A sub-admin who can manage users, pipeline, and map assets
3. **Super Admin:** The owner of the system with full control over the other users and data hosted on the service.

User Stories

1. Match Overview
 - 1.1. As a player, I want to have easy access to all my previous matches so that I can see changes over time.
 - 1.2. As a player, I want to access my previous matches and their statistics so that I can learn from them.
 - 1.3. As a player, I want personalized game-play recommendations so that I can improve my skills.
2. Match Replays
 - 2.1. As a player, I want a frame by frame replay of my match so that I can see matches as a whole.
 - 2.2. As a player, I want frame by frame analysis and recommendations on match replays so that I can see when and where mistakes happened.

3. Player evaluation

- 3.1. As a player, I want a breakdown of my recent performance so that I can see and better understand my current skill level.

4. User Management

- 4.1. As a super admin, I want to assign or demote admin accounts so that I can have full control over who holds elevated privileges.
- 4.2. As a super admin, I want to promote a user to admin so that I can delegate platform management responsibilities
- 4.3. As an admin, I want to deactivate player accounts so that I can revoke access without permanently deleting their data.

5. Data Management

- 5.1. As an admin, I want to view all imported match sessions so that I can verify that data has been ingested correctly.
- 5.2. As an admin, I want to delete a match session so that I can remove duplicate or corrupt data from the system
- 5.3. As an admin, I want to manually trigger a re-ingestion of a match file or other asset so that I can recover from a failed pipeline run

6. System Health

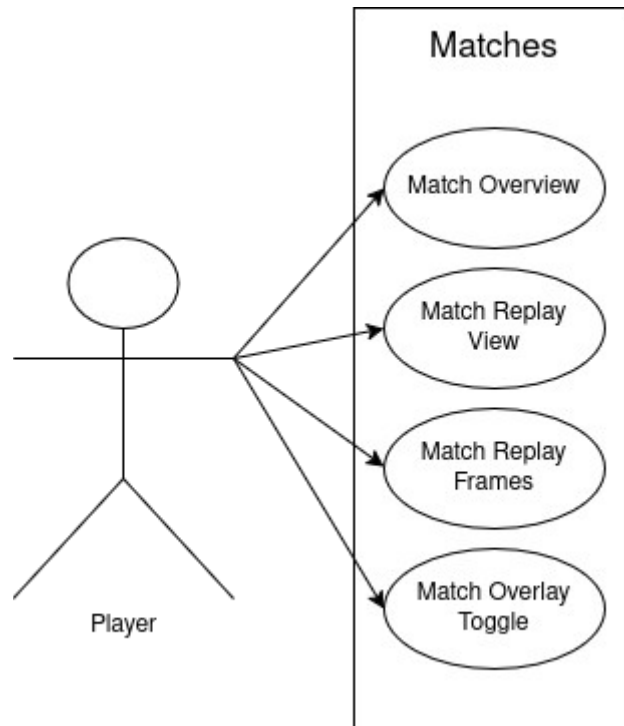
- 6.1. As an admin, I want to view key system metrics so that I can monitor platform health at a glance.
- 6.2. As an admin, I want to see a log of recent system events and error so that I can diagnose issues without accessing the server directly.

7. Register and Login

- 7.1. As a user, I want to be able to make and log in to a secure and safe profile, so that I can have access to the system without concerns for integrity and security of my data

Use Cases

Matches



UC 1. Match Overview

Actor: Player

TUCBW: The player selects a specific match from the match history page

TUCEW: The system pulls match data, processes it, and then displays detailed match information

UC 2. Match Replay View

Actor: Player

TUCBW: The player selects a match on the match replay page to view

TUCEW: The system prepares a frame by frame overview and displays this data on the map from that match

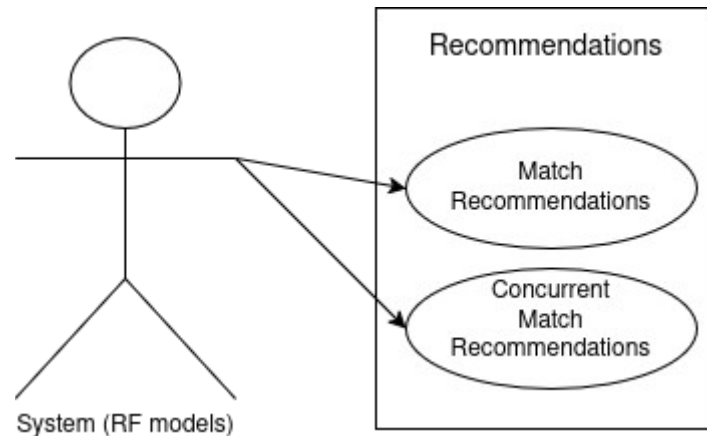
UC 3. Match Overlay Toggle

Actor: Player

TUCBW: The player selects from a list of toggles which elements to display on the overview

TUCEW: The system reloads the map overview to display selected elements without reloading the entire page

Recommendations



UC 4. Match Recommendations

Actor: System (RF models)

TUCBW: The system pulls player data and match information from the static match and runs it as input in the Roles and Champion RF models

TUCEW: The system displays recommended champion choices and team positions for the current user

UC 5. Concurrent Match Recommendations

Actor: System (RF models)

TUCBW: The system pulls frame by frame player data and match information and runs it as input for the Item and Skill RF models

TUCEW: The system displays item purchase and skill upgrade recommendations for the currently displayed frame

Admin Actions

UC 6. Manage User Accounts

Actor: Super Admin

TUCBW: The Super Admin assigns a user profile to a role, “Player” or “Admin”

TUCEW: The system updates the user’s Cognito group membership to reflect the change

UC 7. Error Management

Actor: System

TUCBW: The system saves error messages and logs, and creates an entry for each to display to admin users

TUCEW: The Admin can view, repair, and mark the errors as resolved

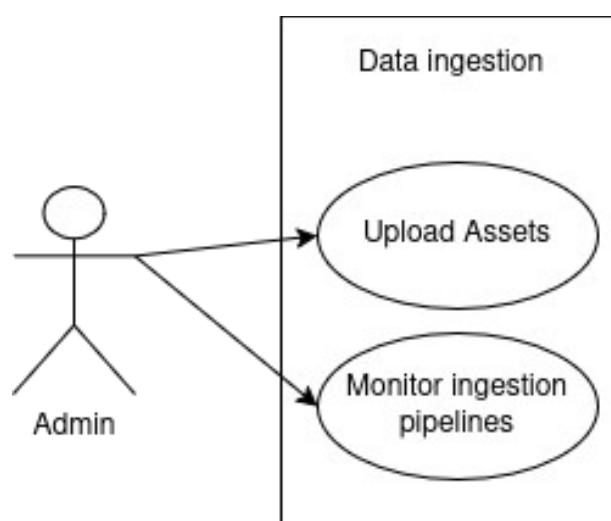
UC 8. Manage Matches

Actor: Super Admin and Admin

TUCBW: The Admin removes any match from the system, given confirmation is received by the system to proceed

TUCEW: The system will confirm the delete and remove the match and all associated coordinate and event data from the database

Data Ingestion



UC 9. Upload Assets

Actor: Admin

TUCBW: The admin uploads a asset file and provides a name and identifier

TUCEW: The system stores the file in AWS S3 and registers it in the PostgreSQL database

UC 10. Monitor Ingestion Pipelines

Actor: Admin

TUCBW: The Admin reviews the error log of an ingestion pipeline and triggers re-ingestion

TUCEW: The system resubmits the process and displays/updates the status thereof in real time

Functional and Non-Functional Requirements

Functional Requirements

FR 1 Sub-system: Match Overview and Replay

FR 1.1 Match History

FR 1.1.1. The system shall display all matches tied to the current user with basic stats

FR 1.1.2. The system shall allow for the filtering of matches

FR 1.1.3. The system shall allow for the searching of a specific match

FR 1.2 Single Match View

FR 1.2.1. The system shall display match details and player statistics within a match for both teams

FR 1.2.2. The system shall display recommendation for the overall match for the current player account (FR 2)

FR 1.3 Match Replay View

FR 1.3.1. The system shall display the map that the match took place in

FR 1.3.2. The system shall allow the user to toggle map overlay elements such as path, deaths, and kills

FR 1.3.3. The system shall allow the user to step through the concurrent match frames and change the overlay concurrently to show the current frames data

FR 1.3.4. The system shall display recommendation for the current frame (FR 2)

FR 1.3.5. The system shall display basic player and team stats for the current frame of the match

FR 2 Sub-system: Recommendation Engine

- FR 2.1 The system shall create KNN and RF models for match recommendation upon system start
- FR 2.2 The KNN and RF models shall be trained on mixed professional E-sports match datasets
 - FR 2.2.1. The system shall ingest data from both winning and losing teams to allow the system to better make recommendations for players in any match
- FR 2.3 The system shall use the input player movement data into the KNN model to recommend more optimal player pathing
- FR 2.4 The system shall run RF models to recommend Team Roles and Champion choices for entire matches (FR 1.2.2)
- FR 2.5 The system shall run RF models to recommend Skill Upgrades and Item Purchases frame-by-frame for a match (FR 1.3.4)
- FR 3 User Profile Page
 - FR 3.1 The system shall display data from the last 20 matches on the user profile page
 - FR 3.2 The system shall display player stats for all matches on the user profile page
 - FR 3.3 The system shall allow a user to customise their interface by toggling between dank and light mode versions
- FR 4 Sub-system: User and Role Management
 - FR 4.1 The Admin page shall display a paginated list of all users retrieved from the AWS Cognito user pool
 - FR 4.2 The system shall allow an admin to assign a user to one of the three Cognito user groups: “Super Admin”, “Admin”, “Player”
 - FR 4.3 The system shall allow an admin to deactivate a Cognito user account without deleting it
 - FR 4.4 The system shall allow an admin to manually create a new user account in Cognito
 - FR 4.5 All role changes shall be enforced at the API layer via JWT claims derived from Cognito group membership
- FR 5 Sub-System: Ingestion Pipeline Monitor
 - FR 5.1 The Admin page shall display a list of ingested jobs with status: Pending, processing, Complete, or Failed
 - FR 5.2 The system shall display the error message for any Failed ingestion job
 - FR 5.3 The system shall allow an admin to manually re-trigger a failed ingestion job
 - FR 5.4 The pipeline status display shall reflect job state changes without requiring a full page reload

FR 6 Sub-system: match Data Management

FR 6.1 The Admin page shall display all matches with metadata

FR 6.2 The system shall allow an admin to delete a match and cascade delete all associated coordinate and event records

FR 6.3 The system shall require explicit confirmation before executing a delete operation

FR 6.4 The system shall allow filtering of matches by date range and map name

FR 7 Asset Management

FR 7.1 The Admin page shall allow upload of asset files (like map images) to AWS S3

FR 7.2 The system shall register uploaded map assets in PostgreSQL with a unique identifier used by the coordinate overlay system

FR 7.3 The admin page shall display all currently available assets with their identifiers

FR 8 User Authentication and Profiles

FR 8.1 The system shall provide secure login and registration through AWS Cognito

FR 8.2 The system shall link user profiles to their Riot Games Ids, allowing for personalized data access and analysis

FR 9 Additional

FR 9.1 The system shall allow super admins to toggle on and off user registration services

FR 9.2 The system shall provide a help/FAQ page for users to access, users are also allowed to give basic feedback on FAQ answers

Non-Functional/ Quality Requirements

1. Security

1. All private user credentials, such as passwords, are managed by AWS Cognito.
2. Users are assigned Cognito groups to enforce access control to certain features.

2. Reliability

1. The system should achieve 98% uptime and be able to recover from critical failure within 5 minutes.

3. Maintainability

1. New features and fixes should be able to stage and deploy within an hour of upload to repository.
2. Unit tests should cover at least 65-70% of the total codebase for staging and deployment.
4. Usability
 1. The system should allow new users to navigate and use core features within the first 2-5 minutes on the site, with 80% user satisfaction during usability testing.
5. Availability
 1. The System should remain online 24/7, allowing for downtime of less than 5 minutes during deployment of updated code.

Domain model

